Determination of Artificial Muscle Placement for Biomimetic Humanoid Robot Legs

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Abstract. Bipedal robotic leg kinematics and dynamics are a key component in designing biomimetic humanoid robots. This work describes the process of designing artificial muscle attachment locations of the legs of a bipedal robot utilizing pneumatic artificial muscles (PAMs). PAMs offer similar force and activation times to real muscles, while being lightweight and low power. However, not all properties are identical, and substituting artificial muscles in with the same attachment locations results in different torque profiles about each joint. This work analyzes muscle length, moment arm, and isometric force of muscles over a range of different joint configurations, based on muscle attachment and wrapping points of the Gait2392 model in OpenSim. These parameters are then used to find joint torques produced by each muscle. This process is repeated for a model of the robot actuated by PAMs. Joint torques for individual muscles and groups of muscles are then compared between the models. Results indicate that several muscles torque profiles in the robot model closely match that of the human model, however, many muscles fall far short of the human capabilities. Where appropriate, additional results demonstrate how muscle attachment locations on the robot have been modified to better match the torque capabilities demonstrated by the OpenSim model. Matlab tools created for this project will facilitate further design refinement before implementation on the robot.

Keywords: Biomechanics, PAMs, Moment Arm, Joint Torque, Humanoid Robot, Biarticular Muscles, Over-Actuated System, OpenSim.

1 Introduction

Biomimetic robots attempt to replicate the anatomical motions and biological systems of living organisms. Of particular interest to humans are biomimetic bipedal humanoid robots. Two different processes are often utilized in developing biomimetic robots. One approach is to engineer the robotic systems to match observed biological principles, while the other works to match the biology and then observe how the system behaves in order to gain insight into the inner workings of biological systems. The work presented here details part of the design process for developing human-mimetic robot legs that falls into the second category [1].

Several human-mimetic robots have been built that fall into this type. These include Lucy [2], Pneumat-BS [3], and the robot built by Zang, et al. [4]. These robots utilize McKibben style Pneumatic Artificial Muscles (PAMs) to actuate the joints. These artificial muscles have many gross properties that mimic those of human muscles [5]. These muscles can only be used to pull, not push. Force-length properties of PAMs have been shown in experiments to be similar to muscles [6] and they can be used in accordance with the Hill muscle model [7]. PAMs are also useful to use as they have a power-to-volume and power-to-weight ratios equal to or greater than other actuators used in robotics [7].

When these robots were developed, many simplifications were used to position artificial muscle attachment locations at the locations of the analogous muscle on the human body. These simplifications reduce the versatility of the robots and their biomimetic similarity to humans. The properties of the PAMs do not match as closely with human muscles as would be most beneficial. Specifically, these artificial muscles cannot contract as much as human muscles: up to 18% in the PAMs vs 38% in biological muscles [8]. As such, if the artificial muscles are positioned at the same locations as the human muscles, joint range of motion, and joint torques are severely limited as compared to humans. Therefore, it is more pertinent to have attachment locations that allow the muscles to match the resultant torque and range of motion profiles that each muscle is able to contribute to movement.

This work compares the torque profiles of multiple muscles for a bipedal robot designed with the same attachment locations as a human. This is done by creating tools in Matlab that accurately reproduce results from the OpenSim [9] benchmark human model Gait2392 [10]. A copy of this model is then made, and muscle properties are changed from human properties to those of the PAMs chosen to be used for the robot. After initial analysis between the robot and OpenSim model, several artificial muscles were removed from the robot model and the attachment points on others have been modified to enable larger joint ranges of motion. Individual muscles are compared, as are groups of muscles based on their anatomical function [11]. Next steps in the bipedal humanoid development process are described.

2 Materials and Methods

For this project, the model Gait 2392 was used from OpenSim. Gait 2392 has 23 degrees of freedom (DoFs), 92 musculotendon actuators to represent 76 real muscles of torso and lower limbs, and 12 rigid-body segments. This model is based on a human subject that is about 1.8 m tall [9]. We first reproduced the segment lengths, muscle attachment locations, and muscle model in Matlab to verify the design process for the robot model and enable more effective comparison of data between the robot and human model.

2.1 Bone Segments

Joint centers in the Matlab model are placed in the same locations with the same orientations as is found in the OpenSim model (see Fig. 1). Using the same bone segment geometry in both the robot model and OpenSim model eliminates the need to perform scaling. Included in the model are details such as the moving knee origin with respect the femur origin. In both models the patella is removed, and the quadriceps insertion location on the patella is modeled in the tibia frame as a function of knee angle.

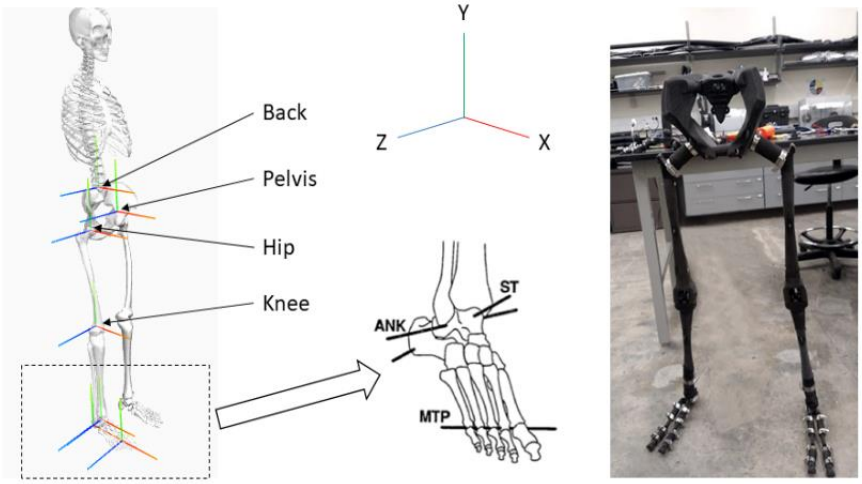


Fig. 1. Joint names and locations. Axis legend shows X axis (red), Y axis (green), and Z axis (blue). Close-up of ankle from Delp, et al. [12]. ANK, ST, and MTP are the ankle, subtalar, and metatarsophalangeal joints, respectively. 3D printed robot skeleton shown on the right.

2.2 Muscle Paths

Muscle attachment locations are also guided by GaitBody2392. Muscle origin and insertion points are kept the same, and the OpenSim moving wrapping points (to guide muscle paths) are replaced by fixed points in the Matlab human model. Several PAM attachment points in the robot model are modified to accommodate the contraction limits of Festo PAMs, and to try to accommodate 3D geometry realities that are not a component of the OpenSim model (i.e. avoiding interference with other actuators). Muscle path routings for GaitBody2392 are shown on the left hand side of Figure 2, below. PAM routing paths for the robot design were exported to OpenSim and are shown on the right hand side for comparison.

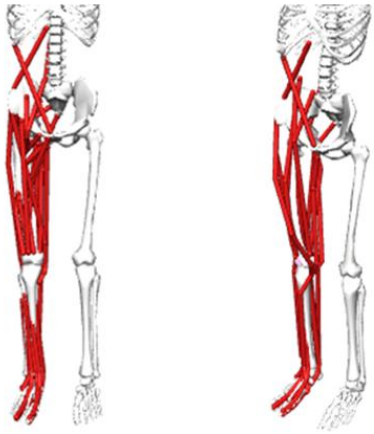


Fig. 2. Skeleton w/ OpenSim muscle paths (left) and PAM paths (right).

2.3 Length and Moment Arm Calculations

All of the skeletal muscles cross at least one joint. The Euclidean norm is used to find the distance between all muscle origins, insertions, and wrapping points for a given joint angle. These norm segments are added up to find the total musculotendon length.

The method to acquire a muscle's moment arm about a joint for a given configuration is described by Hoy [13]. Moment arms are calculated in the body frame.

2.4 PAM Diameter and Resting Length Calculations

The Festo PAMs used for the robot come in three different diameters, 10mm, 20mm, and 40mm [14]. The maximum theoretical force achieved by each actuator scales with diameter size. PAM diameter size for each muscle was chosen to match or exceed the muscle's maximum isometric force specified in the human model. The resting length is found so that it does not exceed maximum length of the muscle over the entire range of motion.

2.5 Force and Torque Calculations

Force used in the torque calculations is the isometric force for a muscle at a certain musculotendon length. This force scales with muscle length according to the force-length curves determined for each muscle. Force calculations for human muscles in the Matlab model were determined from the work of Millard [15], Thelen [16], and Hoy [13]. Details of the model can be found in Thelen [16].

For the Festo PAMs, force vs contraction percentage can be found in the work of Hunt, et al. [17]. For the Festo PAMs, the maximum contraction is a function of the resting muscle length and varies from 14.91% to 17.50% [17].

Torque is calculated once muscle force and moment arm are known (Eq. 1)

$$\text{Torque} = \text{force} * \text{moment arm} \quad (1)$$

The calculations performed in the Matlab human model were checked against OpenSim results generated directly in the program. These are then also compared with the robot model.

3 Results

3.1 Validation of Matlab human model

Figure 4 compares the torque values of the sartorius muscle between OpenSim and the Matlab human model about the hip Z-axis for hip and knee flexion. The maximum difference between the Matlab model and OpenSim was 2%.

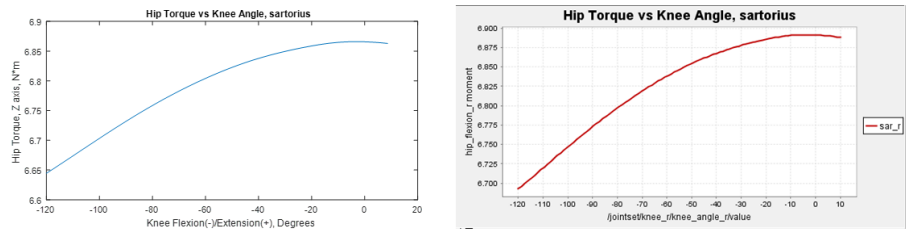


Fig. 3. Comparison of torque values between Matlab human model (left) and OpenSim (right) for hip Z-axis.

3.2 Robot Model

Joint angle limits for human model were determined from range of motion values listed by Platzer [19] and Kapandji [20]. The human model has a maximum hip flexion(+)/extension(-) of +140°/-20°. Hip adduction(+) and abduction (-) is a function of hip flexion, ultimate maximum and minimums are +30°/-80° degrees. Maximum internal hip rotation is 40°. Maximum external hip rotation is -30° with full hip extension and -50° with full hip extension. An initial pass at comparison of joint torques over these ranges of motion determined that even with the ability to change attachment locations, total range of motion would not be achieved due to the limited range of artificial muscle contraction. Therefore, desired joint limits in the robot were reduced. Limits were chosen to include the range of motion required for most human tasks including walking, sitting, and squatting. The robot model is limited to +110°/-15° for flexion(+)/extension(-), and +20°/-60° for ultimate adduction(+)/abduction(-) (Fig. 4). RoM limits for the other joints are listed in Table 1.

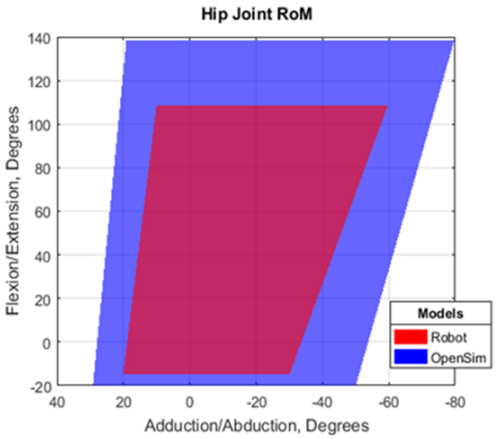


Fig. 4. RoM limits for hip flexion/extension and adduction/abduction for OpenSim (left) and robot (right) models.

Table 1. Joint RoM limits.

Joint	Max	Min	Movement	Axis of Rotation
Knee	10°	-120°	Extension(+)/Flexion(-)	Z
Ankle	20°	-50°	Dorsiflexion(+)/Plantarflexion(-)	Z
Subtalar	25°	-35°	Inversion(+)/Eversion(-)	X
MTP	80°	-30°	Extension(+)/Flexion(-)	Z
Lumbar	20°	-20°	Bending	X
Lumbar	5°	-5°	Rotation	Y
Lumbar	20°	-60°	Extension(+)/Flexion(-)	Z

Table 2 shows the Festo diameter and resting PAM lengths calculated for the robot model's 27 actuators. These values were calculated for the DoF movements listed under the Motion column. Many muscles in the robotic model showed close matching to that of the human model, while several muscles require modifications to muscle attachment points. It is impossible to show the torque results for all muscles in the page limits of this manuscript, therefore only a few select muscles are shown.

Table 2. Resting muscle length and diameter for robot model.

	Resting muscle length, cm			**DoF ranges investigated**
Festo diameter, mm	**10**	**20**	**40**	
Glut Max			64	Hip adduction/abduction and flexion/extension
Semimembranosus			103	Hip and knee flexion/extension
Bicep Femoris, LH		85		Hip and knee flexion/extension
Bicep Femoris, SH		34		Knee flexion/extension
Sartorius	150			Hip and knee flexion/extension

Table 2. Resting muscle length and diameter for robot model.

	Resting muscle length, cm			DoF ranges investigated
Festo diameter, mm	**10**	**20**	**40**	
Adductor Magnus			61	Hip adduction/abduction and flexion/extension
TFL	84			Hip and knee flexion/extension
Gracilis	77			Hip and knee flexion/extension
Iliacus		52		Hip adduction/abduction and flexion/extension
Psoas		53		Hip adduction/abduction and flexion/extension
Rectus Femoris		70		Hip and knee flexion/extension
Vastus Int			41	Knee flexion/extension
Medial Gastro			46	Knee and ankle flexion/extension
Lateral Gastro		48		Knee and ankle flexion/extension
Soleus			36	Knee and ankle flexion/extension
Tibialis Posterior			38	Ankle flexion/extension and foot inversion/eversion
Flex. Digit. Longus	29			Ankle and MTP flexion/extension
Flex. Hallucis Long.	37			Ankle and MTP flexion/extension
Tibialis Anterior		42		Ankle flexion/extension and subtalar inversion/eversion
Peroneus Brevis	38			Ankle flexion/extension and subtalar inversion/eversion
Peroneus Longus		39		Ankle flexion/extension and subtalar inversion/eversion
Peroneus Tertius	40			Ankle flexion/extension and subtalar inversion/eversion
Ext. Digit. Long.	52			Ankle and MTP flexion/extension
Ext. Hallucis Long.	53			Ankle and MTP flexion/extension
Erector Spinae			27	Back flexion and bending
Internal Oblique		35		Back rotation and bending
External Oblique		27		Back rotation and bending

The gastrocnemius is a two-headed biarticular calf muscle modeled as two separate linear actuators in both the human and robot model (Fig. 5). Little modifications have been made to this muscle in the robot as the torque profiles closely match that of the human model. The origin points were moved slightly up the knee to increase the overall length of the muscles. A comparison of the torque over the knee Z-axis for these actuators for different knee and ankle orientations can be seen in Figure 5.

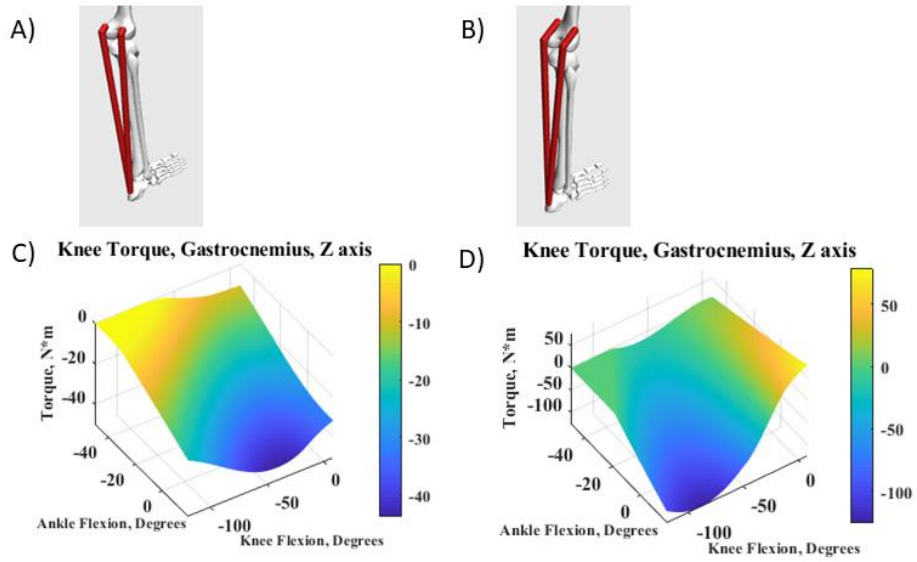


Fig. 5. Gastrocnemius muscle paths (A, B) and knee torque values (C, D) for human (A, C) and robot (B, D) models.

The biceps femoris muscle did not fare nearly as well in comparison. The action of this biarticular muscle is, in part, to perform hip extension and knee flexion. Even after significant lengthening of the muscle (lowering the insertion point on the tibia as shown in Fig. 6), there are still some areas in desired range of motion (maximal knee flexion and hip extension) in which the muscle is not able to produce any force.

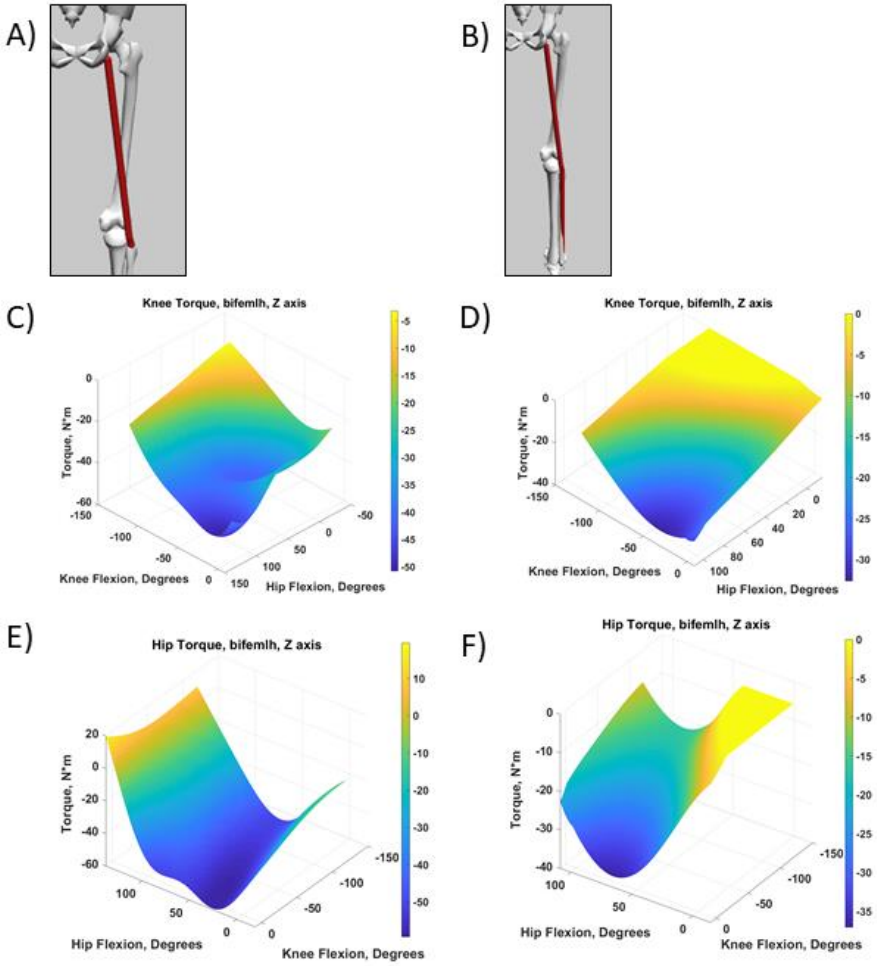


Fig. 6. Biceps Femoris Long Head muscle path (A, B) and torque values (C - F) for human (A, C, E) and robot (B, D, F) models, for knee (C, D) and hip Z-axis (E, F).

Fig. 7 groups and compares the uniarticular muscles of the hip joint (including the psoas). Since many of the gluteus, adductor, and external hip rotators in the model have similar joint torque functions, the number of muscles in the robot model was reduced. Simplification of muscle groups did not result in significant degradation in potential muscle groups. It is important to examine the group as a whole and see if the proposed changes in routings, and muscle consolidation, can provide similar torque. Hip rotation will not affect length as much as the other two hip DoFs and therefore was not investigated. The results demonstrate that the robot should be able to achieve the desired torques in all three degrees of freedom.

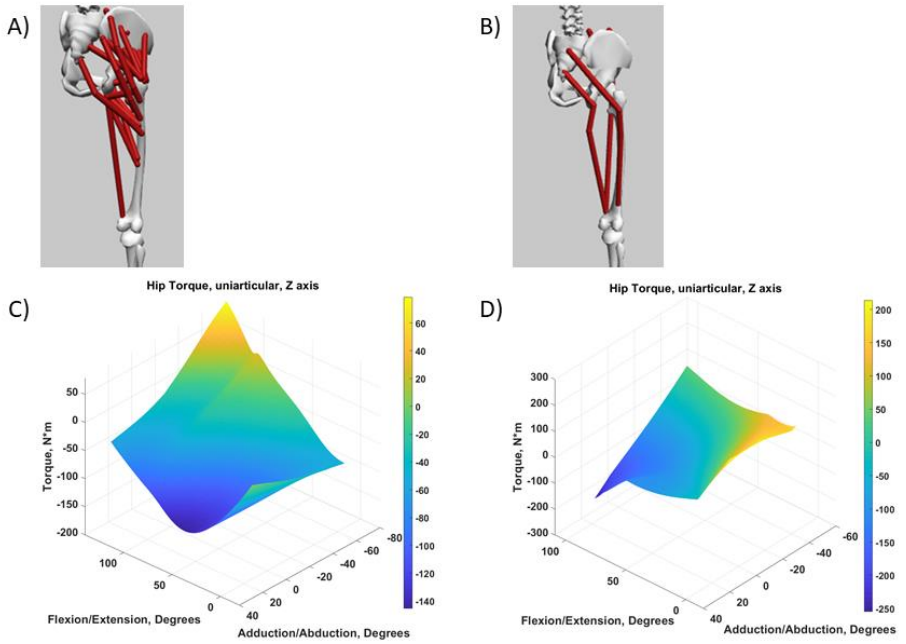


Fig. 7. Uniarticular hip muscle paths (A, B) and hip Z-axis torque values (C, D) for human (A, C) and robot (B, D) models.

4 Discussion

The robot design has 27 pneumatic actuators per side to replace the 46 muscles per side found in GaitBody2392. The muscles need various degrees of refinement. To make this refinement, it is important to look at specific muscles, groups of them based on their function, and groups of all muscles around a particular joint.

The humanoid Matlab model matches with the OpenSim model and enables us to create surface plots of calculated values over 2 degrees of freedom, enabling a more comprehensive view of muscle action than is available in the OpenSim program. This could be expanded to analyzing more than 2 degrees of freedom at once. For example, it could be used to analyze hamstring muscles during hip flexion, abduction, and knee extension movements, though visualizing the data might get difficult.

It is a challenge to achieve maximal joint RoMs and stay within the Festo maximum contraction capabilities. When the artificial muscle is fully contracted, and more range of motion is required, the artificial muscle would need to buckle.

Another limitation of this work is that there is currently little data on Festo hose diameters of 20 mm and 40 mm. These should be tested to see if they do indeed have the same maximum contraction capabilities as the 10 mm Festo muscle. The assumption that their force output will scale based on maximum theoretical force should also be tested. These tests will increase the accuracy of the robot model.

Many individual PAMs and muscle-actuator groups have torque values that have surface plots with significantly different shapes and torque ranges than the OpenSim model. An improvement in the biceps femoris maximum strain was achieved by changing the muscle origin, insertion, and wrapping points, but there are still joint angles in which the muscle is unable to provide any force (Fig. 6). Since this means that the maximum contraction is greater than 17.5%, it might be possible that moving the origin point up the sacrum and better modeling of the 3D geometry might further improve the muscle match. Other muscles that also need to be adjusted in a similar manner include the rectus femoris, psoas, erector spinae, abdominal muscles, calf muscles, and foot muscles.

The gastrocnemius in the robot model can resist much more torque than the one in the human model (Fig. 5), however one problem is that the gastrocnemius in the robot model also has positive torque value for certain ankle and knee position. This is not likely to occur in practice however, and can be eliminated by moving the attachment points out and away from the back of the knee. Solid modelling the skeleton geometry and how the PAMs will wrap around them will aid in this effort.

Looking at the groups of uniarticular hip muscles (Fig. 7), it can be seen that there is not yet a close fidelity between the human model and the robot model. It is also important to note that that the Gait2392 model is not free of errors. For instance, at high degrees of hip flexion or extension some muscle routings pass through other muscles, or pass through bone [10]. Improvements might be made by solid-modeling PAM routings to get more accurate muscle paths or adding back in additional adductor/abductor hip muscles.

Range of motion limitations placed on the robot model helped specify acceptable resting lengths for the PAMs. Hip joint angles for flexion/extension and adduction/abduction were originally set to match the OpenSim model but after experimentation the limits were reduced. Further reducing of joint angle RoM was not as effective in achieving desired lengths when compared to moving the origin, insertion, and wrapping points for muscle actuators.

In all, these tools have been used to calculate and visualize values for length, moment arm, force, and torque for the robot design using PAMs, and they have been highly valuable in determining attachment locations for artificial muscles on a humanoid robot. Values produced in this model can be compared to existing human models. These tools will be effective for further refinement of the robot design. Additionally, more accurate modeling and refinement can be performed through the solid modeled and the final results will be tested with a 3D-printed skeletal structure, enabling us to investigate human control of muscles and movement in a more detailed manner than has yet been possible [1].

5 References

1. Steele, A. G., Hunt, A. J., & Etoundi, A. C.: Biomimetic knee design to improve joint torque and life for bipedal robotics. In: Giuliani, M., Assaf, T., Giannaccini, M. F. (eds.) TAROS 2018, LNAI, vol. 10965, pp. 91-102. Springer, Bristol (2018).

2. Vanderborght, B., Van Ham, R., Verrelst, B., Van Damme, M., Lefeber, D.: Overview of the lucy project: Dynamic stabilization of a biped powered by pneumatic artificial muscles. Advanced Robotics, 22(10), 1027-1051 (2008).
3. Ogawa, K., Narioka, K., Hosoda, K.: (2011, September). Development of whole-body humanoid "Pneumat-BS" with pneumatic musculoskeletal system. In: Amato, N. M. (ed.) IROS 2011, IEEE/RSJ, pp. 4838-4843. IEEE, San Francisco (2011).
4. Zang, X., Liu, Y., Liu, X., Zhao, J.: Design and control of a pneumatic musculoskeletal biped robot. Technology and Health Care 24(s2), S443-S454 (2016).
5. Chou, C. P., Hannaford, B.: Measurement and modeling of McKibben pneumatic artificial muscles. IEEE Transactions on robotics and automation 12(1), 90-102 (1996).
6. Klute, G. K., Czerniecki, J. M., Hannaford, B.: McKibben artificial muscles: pneumatic actuators with biomechanical intelligence. In: 1999 IEEE/ASME International Conference on Advanced Intelligent Mechatronics, pp. 221-226. IEEE, Atlanta (1999).
7. Tondu, B., Zagal, S. D.: McKibben artificial muscle can be in accordance with the Hill skeletal muscle model. In: 1st IEEE/RAS-EMBS International Conference on Biomedical Robotics and Biomechatronics, pp. 714-720. IEEE, Pisa (2006).
8. Gordon, A. M., Huxley, A. F., & Julian, F. J. (1966). The variation in isometric tension with sarcomere length in vertebrate muscle fibres. The Journal of physiology, 184(1), 170-192.
9. Seth, A., Sherman, M., Reinbolt, J. A., Delp, S. L.: OpenSim: a musculoskeletal modeling and simulation framework for in silico investigations and exchange. Procedia Iutam 2, 212-232 (2011).
10. Gait 2392 and 2354 Models. Standfod University, https://simtk-confluence.stanford.edu/display/OpenSim/Gait+2392+and+2354+Models, last accessed 2019/03/22.
11. Gilroy, A., MacPherson, B., Ross, L., Schünke, M., Schulte, E., Schumacher, U.: Atlas of anatomy. 2nd edn. Thieme, New York (2012).
12. Delp, S. L., Loan, J. P., Hoy, M. G., Zajac, F. E., Topp, E. L., Rosen, J. M.: An interactive graphics-based model of the lower extremity to study orthopaedic surgical procedures. IEEE Transactions on biomedical engineering 37(8), 757-767 (1990).
13. Hoy, M. G., Zajac, F. E., Gordon, M. E.: A musculoskeletal model of the human lower extremity: the effect of muscle, tendon, and moment arm on the moment-angle relationship of musculotendon actuators at the hip, knee, and ankle. Journal of biomechanics 23(2), 157-169 (1990).
14. Festo Fluidic Muscle DMSP. https://www.festo.com/cat/en-us_us/data/doc_enus/PDF/US/DMSP_ENUS.PDF, last accessed 2019/3/22.
15. Millard, M., Uchida, T., Seth, A., Delp, S. L.: Flexing computational muscle: modeling and simulation of musculotendon dynamics. Journal of biomechanical engineering 135(2), 021005 (2013).
16. Thelen, D. G.: Adjustment of muscle mechanics model parameters to simulate dynamic contractions in older adults. Journal of biomechanical engineering 125(1), 70-77 (2003).
17. Hunt, A. J., Graber-Tilton, A., Quinn, R. D.: Modeling length effects of braided pneumatic actuators. In: ASME 2017 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, ASME paper no. V05AT08A008. ASME, Cleveland (2017).
18. Platzer, W.: Color atlas of human anatomy, vol. 1: locomotor system. 5th edn. pp. 244–246. Thieme, New York (2004).
19. Kapandji, I. A., Physiology of the joints: vol. 2 lower limb. 6th edn. Churchill Livingstone, London (1986).